Virus
A generic term to cover all malicious code such as trojans, worms, etc

Trojan
An application that is supposed to be useful but is actually malicious code

on most Java and Symbian phones while Windows Mobile comes with Internet Explorer; on Apple's iPhone you have Safari. All of these browsers have vulnerabilities that can be exploited, although not always on the mobile version. Now that people have started transacting on their mobile phones, we will see more instances of data theft than denial of service (DoS) attacks.

BLUE JACKING

The term BluJacking as it turns out is a misnomer. What is commonly believed by the general public is that BlueJacking is a way to take control of another phone using Bluetooth. Giving rise to the wrong belief that the word is an amalgam of Bluetooth and hijacking. This concept in fact is known as BlueSnarfing.

Bluejacking actually is simply sending messages via Bluetooth to other phones. Most commonly people would rename a contact in their phone to something like "You have been Bluejacked" and send it as a Vcard to nearby phones. More mischievous and startling messages would be something like "Hey there cutie, I like your black top". The technique is often used as a guerrilla marketing tool. The term is supposed to have originated when a Malaysian IT consultant used his phone to advertise a particular phone brand. The name was a combination of the words Bluetooth and ajack – his username on an online fan forum. BlueSnarfing on the other hand has a more malicious intent. It is a technique that could be used to copy sensitive data from a victim's phone as well as take complete control of their devices to make calls. This was done by exploiting a security flaw in the Bluetooth standard of earlier phones that has since been rectified.

Attack vectors of the future

In virus parlance, a vector refers to the mode of attack. For example, an infected USB drive is a common vector for desktop viruses. In the case of mobile phones, there are several virus vectors too. And more are likely to emerge as we enter the 3G era. Some of the attack vectors for mobile phones could be:

Peer-to-peer connectivity – this includes Bluetooth and infrared. Bluetooth is used as a vector by most viruses.

Synchronisation with a PC – PCs are more susceptible to viruses. So, if a phone virus has crept in on one and is lying dormant, it can easily get transmitted when the phone is connected to the PC.

Data services – this includes MMS, SMS, WAP and other packet service connectivity. You click a link or download an application from a non-trusted source.

Wireless LAN – this will be the same as getting a virus across the internet.

Telephony – instances of such viruses have not occurred yet, but perhaps some ingenious method can be developed by malicious coders.

Physical media – Commwarrior has been known to infect physical memory media such as SD cards or memory sticks.

How do you protect yourself?

Prevention is the key. Most big names in antivirus software such as Trend Micro, McAfee, FSecure and Kaspersky have security solutions for mobile phones. In fact, some suggest that much of the hype and panic about mobile viruses has been raised by AV companies. So the obvious question is – do you need an antivirus solution for your mobile phone? The answer to this is ambiguous. Firstly, in the case of mobile viruses spread through MMS, they have to go through the network operator's servers. With malware filters, the threat can be curbed from proliferating. However, for viruses spread through Bluetooth, it's best to have protection native to your phone. Most of these antiviruses have trial versions, with a full version starting at $20 upwards to around $50 a year.

Apart from antivirus packages, disinfection tools designed for specific viruses are also available. Such tools can detect and remove a particular type of virus. FSecure had a disinfection tool for both Cabir and CommWarrior. FortiCleanUp Tool from FortiGuard is also another clean up tool that is available free and offers the removal of most known viruses. Other packages such as NetQin and Commander are also available for certain platforms. But the efficacy of these will only be tested when more sophisticated viruses surface. Spybot Search and Destroy is also available for Windows Mobile, Windows CE and Symbian UIQ 2.

Apart from installing antivirus software there are certain simple precautions that you can take. The most basic precaution is to keep your phone in the hidden mode. You obviously lessen the chances of virus attacks if the phone is not visible to worms at all. Activate Bluetooth only when needed. Other Bluetooth precautions include devices pairing in secured areas and choosing a strong PIN. Some of the same rules of the web apply to the mobile world. Be extremely cautious when opening attachments to messages or emails; be it from known or unknown sources. Keep an eye on file extensions; an alarm should always ring in your head when an innocuous extension such as MP3 or JPG asks you for an application install dialogue box. Another rule of thumb that you must follow is – always choose no. Most of the Bluetooth viruses ask you to install a SIS file. In most cases, simply clicking NO would averted the problem all together. Although this may restrict the applications you install on your phone, it's best to stick to applications from trusted sources to lower the risk of infection.

Perhaps manufacturers and the powers that be should take a cue from what happened with desktops and provide for security from the word go, instead for waiting for malicious code to be developed first and then go about countering it. In conclusion we leave you with the boy scouts motto – be prepared.